

```
#for install the numpy library
!pip install numpy
```

Requirement already satisfied: numpy in /usr/local/lib/python3.13/dist-packages (2.1.3)

```
import numpy as np
```

```
li=[1,2,3]
for i in range (len(li)):
    li[i]=li[i]*2
print(li)
```

[2, 4, 6]

```
arr=np.array([1,2,3])
arr*2
```

array([2, 4, 6])

```
#what is the diffrence between numpy array and list?
#list are the python data type which are mutable in nature
#and stores the heterogenous element .and on the list
#the opretions arre costly .for calculating and storing the
#elements..
#numpy array are data the data type which stiore the homogenous
#collection of element which is stored into the memory
#in efficient manner and on this array we can perform the
#matrix opretions very carefully/easily
```

```
#1d array
arr=np.array([1,2,3])
print(arr)
```

[1 2 3]

```
#2d array(matrix)
arr_2d=np.array([[1,2,3],[4,5,6]])
arr_2d
```

array([[1, 2, 3],
 [4, 5, 6]])

```
#nd array(3d array ko tensor v bolte h)
arry_tensor=np.array([[[1,2,3],[4,5,6]]])
print(arry_tensor)
```

[[[1 2 3]
 [4 5 6]]]

```
arry_tensor.ndim
#for dimension of any array of matix we use arr_name.ndim
```

3

```
m=np.array([[1,2,3],[4,5,6],[7,8,9]])
print (m)
```

[[1 2 3]
 [4 5 6]
 [7 8 9]]

```
m.size
```

9

```
m.shape
```

(3, 3)

```
np.eye(3)
```

```
array([[1., 0., 0.],
       [0., 1., 0.],
       [0., 0., 1.]])
```

```
np.zeros((3,2))
```

```
array([[0., 0.],
       [0., 0.],
       [0., 0.]])
```

```
np.ones((3,4))
```

```
array([[1., 1., 1., 1.],
       [1., 1., 1., 1.],
       [1., 1., 1., 1.]])
```

```
#diff b/w shape and size
#shape tell the num of row and num of column
#size total num of elements which are present into the ndim array
arry_tensor.shape
```

```
(1, 2, 3)
```

```
# arange use to generate the elements b/w a range just
#like we have used range fun in python
#syn=np.arange(start,end+1,step)
np.arange(1,10,2)
```

```
array([1, 3, 5, 7, 9])
```

```
#linspace= this fun is used to genrate the num b/w a range .
#but we can control here the num of elements in a range
#syn=np.linspace(start,end,no.of elemnt chaiye)
np.linspace(1,2,6)
```

```
array([1. , 1.2, 1.4, 1.6, 1.8, 2. ])
```

```
Start coding or generate with AI.
```